

THE NEAR-CRITICAL TWO-ROW CONSTRUCTION FOR CATALAN'S CONSTANT: A SELF-CONTAINED PROOF OF THE WORTHINESS $1 - \varepsilon$ THEOREM

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ABSTRACT. We give a short proof of the theorem formalised in the Lean project `lean-catalan-worthiness`: for every $0 \leq k < k^* = 22.3513\dots$, integer combinations of Zudilin's and Nesterenko's rational approximations to Catalan's constant G , divided by the modulus $D_{6n}^2 2^{\lfloor kn \rfloor}$, yield an admissible sequence (q_n, p_n) with $\log |q_n| \geq H(k)n - o(n)$ and $\log |q_n G - p_n| \leq F(k)n + o(n)$, where $F(k) = \frac{\log 2}{2}(k^* - k) \downarrow 0$; in particular its worthiness $\delta = D/(D + F(k))$ exceeds $1 - \varepsilon$. The proof has three steps: a 2-adic cross-divisibility theorem for the two rows (quoted), a congruence lattice of controlled covolume, and a two-dimensional Minkowski argument with an explicit selection between the two successive minima. We explain precisely what the statement does and does not say.

1. WORTHINESS, AND WHAT THE THEOREM SAYS

Definition 1.1 ([1], Def. 1.1–1.2). *Let $\alpha \in \mathbb{R}$. A pair of integer sequences (q_n, p_n) is admissible for α if eventually $q_n \neq 0$ and $q_n \alpha - p_n \neq 0$, and $|q_n| \rightarrow \infty$. Its worthiness is*

$$\delta(q, p) = \liminf_{n \rightarrow \infty} \frac{-\log |\alpha - p_n/q_n|}{\log |q_n|} = 1 - \limsup_{n \rightarrow \infty} \frac{\log |q_n \alpha - p_n|}{\log |q_n|}.$$

Three facts calibrate the scale. (i) For *every* real α , rounding $p_n = \lfloor n\alpha \rfloor$ (adjusted by one if the form vanishes) gives an admissible sequence with $\delta \geq 1$. (ii) If $\alpha = a/b$ is rational, every admissible sequence has $|q_n \alpha - p_n| \geq 1/b$, hence $\delta \leq 1$. (iii) Consequently an admissible sequence with $\delta > 1$ proves $\alpha \notin \mathbb{Q}$, and $\delta = 1$ is the *rationality ceiling*. Dirichlet's theorem gives $\delta = 2$ for irrational α .

So a worthiness below 1 is never a Diophantine statement about α ; it measures the quality of a *construction*. Zudilin's explicit row for G [3] has $\delta = 0.9025\dots$, Nesterenko's [2] less; the theorem below says that combining the two rows 2-adically produces structured sequences whose worthiness approaches the ceiling. The content is in the rates: the cleared linear forms grow like $e^{F(k)n}$ with $F(k) \rightarrow 0^+$ as $k \rightarrow k^*$; for $k > k^*$ the same lattice contains vectors with $F < 0$, which would prove irrationality if their forms were known to be nonzero (see §6). The Lean statement `catalan_worthiness_one_sub_eps` does not constrain the witnesses and is therefore also provable by rounding; the formalised proof discharges it with the explicit construction, and that construction is what we prove here.

2. INPUTS: THE TWO ROWS

Throughout $D_N = \text{lcm}(1, \dots, N)$, $S_n := D_{6n}^2$, $\log D_N = N + o(N)$, and $\varphi = (1 + \sqrt{5})/2$.

Row 1 (Zudilin). Zudilin's second-order recurrence [3] produces rationals Q_m, P_m ($Q_0 = 1, Q_1 = \frac{7}{4}, P_0 = 0, P_1 = \frac{13}{8}$) with $P_m/Q_m \rightarrow G$, denominators of Q_m pure powers of 2, $2^{e_m} Q_m \in$

$\mathbb{Z}$ and $2^{e_m} D_{2m-1}^2 P_m \in \mathbb{Z}$ with $e_m = \min(6m, 4m + 3 + \lfloor \log_2(2m - 1) \rfloor)$, and rates $\frac{1}{m} \log |Q_m| \rightarrow 5 \log \varphi$, $\frac{1}{m} \log |Q_m G - P_m| \rightarrow -5 \log \varphi$. At the 3-sectioned index $m = 3n$ we set

$$X_{1,n} := S_n 2^{e_{3n}} Q_{3n}, \quad Y_{1,n} := S_n 2^{e_{3n}} P_{3n} \quad (\in \mathbb{Z}),$$

with $\log |X_{1,n}| = A_Z n + o(n)$, $\log |X_{1,n} G - Y_{1,n}| = E_Z n + o(n)$, $A_Z = 12 + 12 \log 2 + 15 \log \varphi$, $E_Z = 12 + 12 \log 2 - 15 \log \varphi$.

Row 2 (Nesterenko). Nesterenko's (4, 7) construction [2] gives rationals B_n, C_n with $J_n := 4B_n G - C_n > 0$, $J_n \rightarrow 0$, such that $V_n := 4^{7n+1} S_n B_n$ and $U_n := 4^{7n} S_n C_n$ are integers. Set $X_{2,n} := V_n$, $Y_{2,n} := U_n$, so $X_{2,n} G - Y_{2,n} = 4^{7n} S_n J_n$, with rates $A_N = 12 + 14 \log 2 + 2 \log T^+$, $E_N = 12 + 14 \log 2 + 2 \log T^-$, where $T^\pm = (3303 \pm 437\sqrt{57})/144$, $T^+ T^- = 32/27$. Numerically $A_N = 35.37$, $E_N = 14.39$, $D := A_N - E_N = 2 \log(T^+/T^-) = 14.9616$.

The crossed gap. $A_Z + E_N < A_N + E_Z$, equivalently $30 \log \varphi = 14.436 < D$. [[1],
crossed_gap_NC]

The 2-adic package. Write $X_{i,n} = S_n a_{i,n}$. Then for all $n \geq 1$

$$(1) \quad 2^{24n} \mid a_{1,n} Y_{2,n} - a_{2,n} Y_{1,n}.$$

This is the one input whose proof we do not reproduce: it is proved in [1] (`NesterenkoCross.lean`, `dvd_reduced_cross_N`) from the 2-adic structure of the two rows — Nesterenko's linear form J_n is, 2-adically, a unit times an explicit ${}_3F_2(1)$ factor whose terms have 2-adic norm controlled by half-integral Pochhammer symbols, and the cross determinant inherits 2^{24n} exactly. (It is the statement that the two rows realise the same extension class and converge 2-adically to the same limit at the same rate; cf. the discussion of Theorem F in the companion paper.)

3. THE CONGRUENCE LATTICE

Fix $0 \leq k \leq 24$ and put $T_n := 2^{\lfloor kn \rfloor}$, $M_n := S_n T_n$, so $\log M_n = \sigma(k)n + o(n)$ with $\sigma(k) = 12 + k \log 2$. By (1), $T_n \mid a_{1,n} Y_{2,n} - a_{2,n} Y_{1,n}$, hence

$$(2) \quad M_n \mid \Delta_n := X_{1,n} Y_{2,n} - X_{2,n} Y_{1,n} = S_n (a_{1,n} Y_{2,n} - a_{2,n} Y_{1,n}).$$

Moreover $\Delta_n = X_{2,n} L_{1,n} - X_{1,n} L_{2,n}$ with $L_{i,n} := X_{i,n} G - Y_{i,n}$, and the crossed gap gives $|X_{2,n} L_{1,n}| = e^{(A_N + E_Z)n + o(n)} \gg |X_{1,n} L_{2,n}|$, so $\Delta_n \neq 0$ and $\log |\Delta_n| = (A_N + E_Z)n + o(n)$.

Let $V_n := \frac{1}{M_n} (\mathbb{Z}(X_{1,n}, Y_{1,n}) + \mathbb{Z}(X_{2,n}, Y_{2,n})) \subset \mathbb{Q}^2$, a lattice of covolume $|\Delta_n|/M_n^2$, and $\Gamma_n := V_n \cap \mathbb{Z}^2$: the integer pairs (q, p) of the form $(\frac{c_1 X_1 + c_2 X_2}{M}, \frac{c_1 Y_1 + c_2 Y_2}{M})$.

Lemma 3.1. $[V_n : \Gamma_n] \leq M_n$; hence $\text{covol } \Gamma_n \leq |\Delta_n|/M_n$.

Proof. $[V_n : \Gamma_n]$ is the order of the image A of V_n in $(\mathbb{Q}/\mathbb{Z})^2$, generated by $v_i = (X_i, Y_i)/M$. The first-coordinate projection $\pi(A) = \langle X_1/M, X_2/M \rangle \subset \mathbb{Q}/\mathbb{Z}$ has order M/g , $g := \gcd(M, X_1, X_2)$. If $c_1 v_1 + c_2 v_2 \in \ker \pi$, i.e. $c_1 X_1 + c_2 X_2 = Mt$, then $X_1(c_1 Y_1 + c_2 Y_2) = Y_1 Mt + c_2 \Delta \equiv 0 \pmod{M}$ by (2), and likewise $X_2(c_1 Y_1 + c_2 Y_2) \equiv 0$; so $M \mid g \cdot (c_1 Y_1 + c_2 Y_2)$, and the second coordinate of the element lies in $\frac{g}{M} \mathbb{Z}/\mathbb{Z}$, a group of order g . Thus $|A| \leq (M/g) \cdot g = M$. $\square$

Since $\Gamma_n \subset V_n$, $\text{covol } \Gamma_n = (|\Delta_n|/M_n^2)[V_n : \Gamma_n] \in [|\Delta_n|/M_n^2, |\Delta_n|/M_n]$; a sublattice $\Gamma'_n \subset \Gamma_n$ of index $j = \lceil (|\Delta_n|/M_n) / \text{covol } \Gamma_n \rceil$ (e.g. $\{(q, p) \in \Gamma_n : j \mid q\}$) then has

$$(3) \quad \mathcal{D}_n := \text{covol } \Gamma'_n \in [|\Delta_n|/M_n, 2|\Delta_n|/M_n], \quad \log \mathcal{D}_n = (A_N + E_Z - \sigma(k))n + o(n).$$

Every element of Γ'_n is an integer pair (q, p) ; the map $(q, p) \mapsto (q, \ell)$, $\ell := qG - p$, is unimodular, so the image $\Lambda_n \subset \mathbb{R}^2$ has the same covolume $\mathcal{D}_n$.

4. MINKOWSKI: THE BALANCED PAIR

Set $x := \frac{1}{2}(\sigma(k) + E_N - E_Z)$ and

$$H := A_N - x, \quad F := x + E_Z - \sigma(k) = \frac{1}{2}(E_Z + E_N - \sigma(k)) = \frac{\log 2}{2}(k^* - k), \quad k^* := \frac{E_Z + E_N - 12}{\log 2},$$

so that $H + F = A_N + E_Z - \sigma(k)$ is the rate of $\mathcal{D}_n$, $H - F = D > 0$, and $F > 0$ iff $k < k^*$. Numerically $k^* = 22.3513\dots$ and $\delta(k) := 1 - F/H = D/(D + F(k))$; $\delta(22) = 0.99193\dots$

Apply Minkowski's second theorem in the plane to Λ_n and the box $K = [-U, U] \times [-V, V]$ with $U := e^{Hn}$, $V := 2\mathcal{D}_n/U$: $\lambda_1\lambda_2 \text{ vol } K \leq 4\mathcal{D}_n$ gives $\lambda_1\lambda_2 \leq \frac{1}{2}$. Let $b_1 = (Q_1, \ell_1)$, $b_2 = (Q_2, \ell_2)$ attain $\lambda_1 =: \mu \leq 1$ and $\lambda_2 \leq 1/(2\mu)$; in dimension two they form a basis of Λ_n , so $|Q_1\ell_2 - Q_2\ell_1| = \mathcal{D}_n$, and since $Q_i\ell_j - Q_j\ell_i = Q_jP_i - Q_iP_j$ this is an integer. Writing $\mu = e^{-rn}$ ($r \geq 0$) and using (3), there is $\epsilon_n \rightarrow 0$ (with $\sqrt{\epsilon_n}n \rightarrow \infty$, e.g. after replacing ϵ_n by $\max(\epsilon_n, n^{-1})$) such that

$$(4) \quad |Q_1| \leq e^{(H-r+\epsilon_n)n}, \quad |\ell_1| \leq e^{(F-r+\epsilon_n)n}, \quad |\ell_2| \leq e^{(F+r+\epsilon_n)n}, \quad |Q_1P_2 - Q_2P_1| \geq e^{(H+F-\epsilon_n)n}.$$

(These four inequalities, with $r = r_n \geq 0$ arbitrary, are the only thing used below.)

5. THE SELECTION

Lemma 5.1 (one step). *Let $0 < F < H$, $0 < \epsilon \leq \frac{1}{9}$, n with $e^{\sqrt{\epsilon}n} \geq 3$ and $D_n := (H - \sqrt{\epsilon} - 2\epsilon)n - \log 2 > 0$, and let $(Q_i, P_i) \in \mathbb{Z}^2$, $\ell_i = Q_iG - P_i$, satisfy (4) for some $r \geq 0$. Then there is an integer pair (q, p) , an integer combination of $(Q_1, P_1), (Q_2, P_2)$, with*

$$q \neq 0, \quad \ell := qG - p \neq 0, \quad \log |q| \geq D_n, \quad \frac{\log |\ell|}{\log |q|} \leq \frac{N_n}{D_n}, \quad N_n := (F + \sqrt{\epsilon} + \epsilon)n + \log \frac{3}{2}.$$

Proof. Write $\Delta := Q_1P_2 - Q_2P_1$, so $Q_1\ell_2 - Q_2\ell_1 = -\Delta \neq 0$; in particular ℓ_1, ℓ_2 do not both vanish. If $\log |\ell| \leq 0$ the ratio condition is automatic, so we only need it when $|\ell| > 1$.

Case C: $\ell_1 = 0$. Then $Q_1 \neq 0$ and $\ell_2 \neq 0$. Take $(q, p) = (Q_2 + tQ_1, P_2 + tP_1)$ with $t \in \mathbb{Z}$ of the sign of Q_1Q_2 (or any sign if $Q_2 = 0$) and $|t|$ so large that $|q| \geq |t||Q_1| - |Q_2| \geq \exp \max(D_n, \log |\ell_2| \cdot D_n/N_n)$. Then $\ell = \ell_2 \neq 0$ and both conditions hold.

Case B: $\ell_1 \neq 0$, $r > \sqrt{\epsilon}$. Choose $m \in \mathbb{Z}$ with $0 < |\ell_2 - m\ell_1| \leq \frac{3}{2}|\ell_1|$ ($m = \lfloor \ell_2/\ell_1 \rfloor$, or $m + 1$ if that gives 0). Put $(q, p) = (Q_2 - mQ_1, P_2 - mP_1)$, $\ell = \ell_2 - m\ell_1$. The identity $q\ell_1 - Q_1\ell = -\Delta$ gives

$$|q||\ell_1| \geq |\Delta| - |Q_1||\ell| \geq e^{(H+F-\epsilon)n} - \frac{3}{2}e^{(H-r+\epsilon)n}e^{(F-r+\epsilon)n} = e^{(H+F-\epsilon)n}(1 - \frac{3}{2}e^{(-2r+3\epsilon)n}) \geq \frac{1}{2}e^{(H+F-\epsilon)n},$$

because $-2r+3\epsilon < -\sqrt{\epsilon}$ (as $r > \sqrt{\epsilon}$ and $3\epsilon \leq \sqrt{\epsilon}$) and $e^{-\sqrt{\epsilon}n} \leq \frac{1}{3}$. Hence $|q| \geq e^{(H+F-\epsilon)n}/(2|\ell_1|) \geq \frac{1}{2}e^{(H+r-2\epsilon)n} \geq \frac{1}{2}e^{(H-2\epsilon)n}$, so $\log |q| \geq D_n$, and $|\ell| \leq \frac{3}{2}e^{(F-r+\epsilon)n} \leq \frac{3}{2}e^{(F+\epsilon)n}$, so $\log |\ell| \leq N_n$.

Case A: $\ell_1 \neq 0$, $r \leq \sqrt{\epsilon}$. From $|\Delta| \leq |Q_1||\ell_2| + |Q_2||\ell_1|$ one of the two products is $\geq |\Delta|/2$. If $|Q_1||\ell_2| \geq |\Delta|/2$: then $|Q_1| \geq e^{(H+F-\epsilon)n}/(2e^{(F+r+\epsilon)n}) \geq \frac{1}{2}e^{(H-\sqrt{\epsilon}-2\epsilon)n}$; take $(q, p) = (Q_1, P_1)$, $\ell = \ell_1 \neq 0$, $|\ell_1| \leq e^{(F+\epsilon)n}$. If $|Q_2||\ell_1| \geq |\Delta|/2$: then $|Q_2| \geq e^{(H+F-\epsilon)n}/(2e^{(F-r+\epsilon)n}) \geq \frac{1}{2}e^{(H-2\epsilon)n}$. If $\ell_2 \neq 0$ take $(q, p) = (Q_2, P_2)$, $|\ell_2| \leq e^{(F+\sqrt{\epsilon}+\epsilon)n}$. If $\ell_2 = 0$ take $(q, p) = (Q_2 + sQ_1, P_2 + sP_1)$ with $s = \pm 1$ chosen so that $|Q_2 + sQ_1| \geq |Q_2|$; then $\ell = s\ell_1 \neq 0$ and $|\ell| \leq e^{(F+\epsilon)n}$. In all sub-cases $\log |q| \geq D_n$ and $\log |\ell| \leq N_n$. $\square$

6. ASSEMBLY

Theorem 6.1. *Let $0 \leq k < k^*$. There is an admissible sequence (q_n, p_n) for G , each (q_n, p_n) an integer combination of $(X_{1,n}, Y_{1,n})$ and $(X_{2,n}, Y_{2,n})$ divided by $M_n = D_{6n}^2 2^{\lfloor kn \rfloor}$, such that*

$$\log |q_n| \geq H(k)n - o(n), \quad \log |q_nG - p_n| \leq F(k)n + o(n), \quad F(k) = \frac{\log 2}{2}(k^* - k), \quad H(k) = D + F(k).$$

Consequently $\delta(q, p) \geq 1 - F(k)/H(k) = D/(D + F(k))$, which exceeds $1 - \varepsilon$ once $k^* - k < \frac{2\varepsilon D}{(1-\varepsilon)\log 2}$; and $\delta(22) > 0.99$.

Proof. For each large n apply Lemma 5.1 with $H = H(k)$, $F = F(k) > 0$, $\epsilon = \epsilon_n$ from (4); the hypotheses $e^{\sqrt{\epsilon_n}n} \geq 3$ and $D_n > 0$ hold for large n since $\sqrt{\epsilon_n}n \rightarrow \infty$ and $\epsilon_n \rightarrow 0$. The output (q_n, p_n) lies in $\Gamma'_n \subset \Gamma_n$, hence is of the stated form, with $q_n \neq 0$, $q_n G - p_n \neq 0$, $\log |q_n| \geq D_n \rightarrow \infty$ (so $|q_n| \rightarrow \infty$: admissible), and $\log |q_n G - p_n| \leq \max(0, N_n)$. Since $D_n = (H - o(1))n$ and $N_n = (F + o(1))n$, the rates follow, and $\limsup \log |q_n G - p_n| / \log |q_n| \leq \limsup N_n / D_n = F/H$ gives $\delta \geq 1 - F/H$. The numerical values: $E_Z + E_N - 12 = 15.4928$, $k^* = 22.3513$, $D = 14.9616$, $F(22) = 0.12176$, $\delta(22) = 0.99193$. $\square$

7. WHAT THE THEOREM DOES NOT SAY, AND WHY k^* IS WHERE IT STOPS

Remark 7.1 (the ceiling). For $k > k^*$ one has $F(k) < 0$, and (4) still produces, for each n , a lattice vector b_1 with $|q| \leq e^{(H-r)n}$ and $|qG - p| \leq e^{(F-r)n} \rightarrow 0$. If its form were nonzero for infinitely many n , G would be irrational. But now $UV = 2D_n$ with $V < 1$, the box has area comparable to the covolume, and the selection degenerates to Dirichlet's theorem: nothing prevents b_1 from being the *kernel vector*: for every rational $G = a/b$ the lattice Γ'_n contains a pair $(q, p) \propto (b, a)$ (explicitly $(q, p) = \pm(h_n/T_n)(b, a)$ with h_n the cross determinant, cf. P2_STRUCTURE.md), whose form is 0. Lemma 5.1 uses $F > 0$ precisely to rule this out (in Case C the kernel vector is harmless only because $F > 0$ lets us inflate q). The hypothesis $F > 0$ is the Lean hypothesis `hF`, unsatisfiable for $k \geq k^*$. Replacing the lattice's first minimum by its first minimum *in the positive cone* (both rows' forms have known sign) would restore nonvanishing; that this loses only $e^{o(n)}$ is Lemma P2 of the companion paper, which is equivalent to $G \notin \mathbb{Q}$ and, by the continued-fraction analysis of P2_STRUCTURE.md, amounts to the parity of a convergent index whose determination costs 14.4 nats of G per unit n .

Remark 7.2 (rational control). The construction never uses more than $\sim n$ digits of G : replacing G by $G^* = \text{bestappr}(G, 10^{320})$ reproduces every table to $n \approx 100$. This is consistent with §1: a worthiness < 1 cannot distinguish G from a rational number.

Remark 7.3 (where the 2-adic input acts). Without (1) one could only take $k = 0$ (modulus S_n alone), giving $F(0) = \frac{\log 2}{2}k^* = 7.75$ and $\delta(0) = 0.659$. The package (1) is what lets the modulus absorb 2^{kn} up to $k^* = 22.35 < 24$; the $24n$ is not used in full because the archimedean rates, not the 2-adic divisibility, set k^* . The earlier construction of [1] pairing Zudilin's row with Zagier's modular row **E** gives $\delta = 0.857914$ by the same argument with $(A_E, E_E) = (12 + 18 \log 2, 12 + 12 \log 2)$ and $k = 24$.

Provenance. The theorem and every step above are formalised in [1] (`NearCriticalAssembly.lean`, `GeometrySelection.lean`, `TwoRow.lean`); Lemma 3.1 replaces the project's explicit “thinned basis” by an index argument. The 2-adic package (1) is the only input not reproved here.

REFERENCES

- [1] *lean-catalan-worthiness*, Lean 4/Mathlib formalisation (Aristotle/Harmonic and the authors), 2026; exported theorem `Catalan.catalan_worthiness_one_sub_eps_nesterenko`.
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